

QoE-aware Download Control and Bitrate Adaptation for Short Video Streaming

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ABSTRACT

Nowadays, although the rapidly growing demand for short video sharing has brought enormous commercial value, considerable bandwidth usage becomes a problem for service providers. To save costs of service providers, the short video applications face a critical conflict between maximizing the user quality of experience (QoE) and minimizing the bandwidth usage. Most of existing bitrate adaptation methods are designed for the livecast and video-on-demand instead of short video applications. In this paper, we propose a QoE-aware adaptive download control mechanism to ensure the user QoE and save the bandwidth, which can download the appropriate video according to user retention probabilities and network conditions, and pause the download when the buffered data is enough. The extensive simulation results demonstrate the superiority of our proposed mechanism over the other baseline methods.

CCS CONCEPTS

• **Networks** → **Application layer protocols**; • **Information systems** → **Multimedia streaming**.

KEYWORDS

Short video, Download control, Bitrate adaptation

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1 INTRODUCTION

Short video streaming is a new form of video that is shared on online social platforms based on user-generated content [9]. In recent years, short video media emerges quickly and becomes one of the most popular multimedia applications on Internet, which drives the rapid development of short-video sharing platforms such

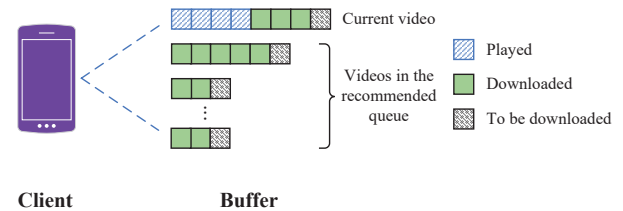


Figure 1: The download model in short video applications.

as Tik-Tok, Kuaishou, Kwai, and so on. In early 2020, the numbers of daily active users of Douyin and Kuaishou in China were already over 400 and 300 millions, respectively [6, 10].

As shown in Figure 1, we consider the following download model for the short video applications. The user can slide away anytime while watching the current video and continue to watch the next video in a recommended queue. The client can download the current video or preload the videos in the recommended queue to guarantee the user quality of experience (QoE).

However, if the user slides away, the downloaded but unwatched data does not contribute to the user QoE, which results in wasting bandwidth. To reduce the bandwidth wastage, the client should select a suitable video chunk and its bitrate based on user viewing behaviors and network conditions. Unfortunately, saving bandwidth and improving user QoE are two conflicting goals. On the one hand, although many existing works try to characterize user behaviors in livecast and video-on-demand [5, 7], the download data volume for a certain video is difficult to determine since the short video user's viewing behavior varies greatly. This characteristic of short videos requires the client to consider the download sequence of video chunks instead of simply downloading the current video. On the other hand, selecting the right bitrate for each video chunk also can be very challenging under dynamic network conditions. Today's commercial video applications often use bitrate adaptation schemes that are designed for livecast and video-on-demand [1], such as buffer-based approach [3] and FastMPC [8]. In contrast, short video applications lack customized download control and bitrate adaptation mechanisms [2], which is a Grand Challenge in the ACM Multimedia 2022–Short Video Stream¹.

In this paper, we formulate the video download control problem in short video applications (§2). To address this challenge, we design a download control mechanism (§3) for three critical tasks of the video download process: 1) decide the video to download next; 2) decide the video bitrate for the next-to-download chunk; 3) decide the pausing time to stop downloading whenever appropriate. Our mechanism selects candidate video chunks based on the bandwidth

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¹<https://www.aitrans.online/MMGC2022/>

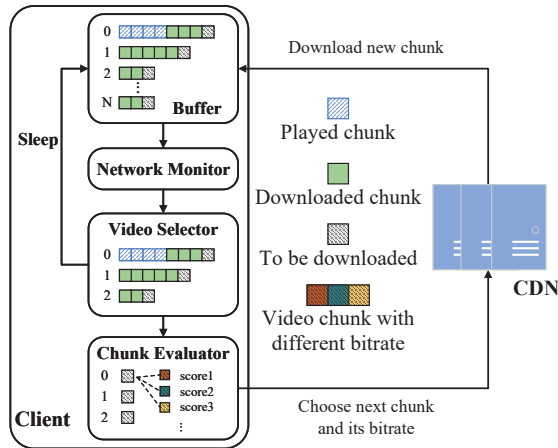


Figure 2: The workflow of our video download mechanism.

estimations and user retention probabilities. The next-to-download video chunk and its bitrate will be selected by calculating a quality score for each bitrate of every candidate video chunk.

We evaluate the performance of the proposed video download control mechanism under various bandwidth conditions (§4). Using trace-driven emulations, we verify that our mechanism consistently outperforms the baseline methods that are used in livecast and video-on-demand on improving QoE and saving bandwidth. Finally, the performance of our mechanism ranked 2nd in ACM Multimedia 2022 Grand Challenge–Short Video Streaming.

2 OVERVIEW

2.1 Problem Formulation

In short video applications, the download performance can be evaluated from the following four metrics: 1) video bitrate, 2) rebuffer time, 3) video smooth (bitrate variation between consecutive video chunks), and 4) bandwidth usage. The user QoE can be given as

$$QoE_i = \sum_j^{M_i} (w_1 \cdot \text{bitrate}_{i,j} - w_2 \cdot \text{rebuffer}_{i,j} - w_3 \cdot \text{smooth}_{i,j}), \quad (1)$$

where i denotes the video index and M_i is the number of chunks in video i that the user has watched. w_1 , w_2 , and w_3 are the preference coefficients. The overall performance can be calculated as

$$Score = \sum_i^N (QoE_i - w_4 \cdot bandwidth_usage_i), \quad (2)$$

where N is the number of videos and w_4 is a coefficient of bandwidth usage. Note that the organizer of the grand challenge sets w_1 , w_2 , w_3 , and w_4 as 1, 1.85, 1, 0.5, respectively.

In this paper, we design a QoE-aware video download control mechanism to obtain the best user QoE while using as little bandwidth as possible by maximizing *Score*.

2.2 Download Control Workflow

As shown in Figure 2, our video download mechanism can control the video chunk download by three major components: 1) network monitor, 2) video selector, and 3) chunk evaluator. The client first downloads the initial video chunk and the network monitor predicts the bandwidth based on the last few downloads. The video selector

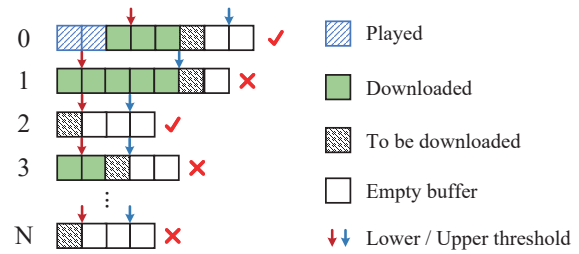


Figure 3: An example of selecting videos by the video selector.

selects the videos that can be chose to download based on the network condition provided by the network monitor. When all the videos have enough number of unplayed buffered chunks, i.e., the network bandwidth resources are sufficient, the client will pause downloading to avoid wasting bandwidth. Otherwise, the chunk evaluator will evaluate each bitrate for every chunk to select the best chunk and its bitrate to download.

3 MECHANISM DESIGN

We design a QoE-aware adaptive download control mechanism to select the video chunk and its bitrate to download next. In this section, we introduce the three major components of our mechanism.

3.1 Network Monitor

The network monitor predicts the bandwidth while using the exponential smoothing method [4], which is pure a statistical-based method. We define the last video chunk size as b_{i-1} when the client downloads the i th chunk and define the download time of the last chunk as t_{i-1} . The average bandwidth of last download can be calculated as $B_{last} = b_{i-1}/t_{i-1}$. The bandwidth B_i can be predicted by B_{last} and the previous bandwidth estimation B_{i-1} as follows

$$B_i = \alpha \cdot B_{last} + (1 - \alpha) \cdot B_{i-1}, \quad (3)$$

where α is a constant less than 1 to indicate the degree of dependence on historical information.

3.2 Video Selector

The video selector selects video chunks from the current video and the recommended queue as the candidates that are considered to download next. We proposed two principles for selecting the videos. 1) If the unplayed buffered amount of a video is less than the lower threshold, all the videos after this video in the recommended queue will not be selected. 2) If the unplayed buffered amount of a video is larger than the upper threshold, this video will not be selected. The video selector selects videos subject to these principles and the chunk evaluator will then select the video chunk and its bitrate. Figure 3 shows an example of selecting videos by the video selector. The selector removes video 1 based on principle 2, removes video 3, 4, ..., N based on principle 1, and thus selects video 0 and 2.

To adapt to the dynamic network bandwidth, we design the adaptive lower and upper thresholds based on the video chunk size and the bandwidth estimation of the network monitor. The lower threshold is designed to disregard the videos after the video that has too little unplayed buffered data, which is calculated as follows

$$threshold_k^{(low)} = \begin{cases} \max(C_1, T_k) & k = 0; \\ C_1 & \text{otherwise;} \end{cases} \quad (4)$$

Table 1: An overview of bandwidth datasets in different network conditions.

	AVG	SD	Max	Min
low bandwidth	1.23	0.52	4.21	0
medium bandwidth	1.71	0.83	5.34	0
high bandwidth	3.55	1.83	14.42	0

Table 2: Parameter settings of our mechanism in evaluation

N	$C_1/C_2/C_3$ (ms)	α	t_{sleep} (ms)	B_{min}/B_{max} (Mbit/s)
7	2000/1000/3000	0.1	500	1.0/3.5

where k indicates the video index ($k = 0$ indicates the current video) and C_1 is a constant to guarantee the effectiveness of the lower threshold. $T_k = b_{i+1}^{(max)}/B_i$ is the maximum estimated download time for the next chunk of current video, where $b_{i+1}^{(max)}$ is the next video chunk size of current video subject to the max bitrate. We use T_k to dynamically adjust the lower threshold when the next chunk download is too slow. The upper threshold is designed to avoid bandwidth waste as follows

$$threshold_k^{(up)} = \begin{cases} C_2 + p_k \cdot \min_{j \in \{1, \dots, N\}} (T_j) & k = 0; \\ p_k \cdot C_3 & \text{otherwise;} \end{cases} \quad (5)$$

where C_2 and C_3 are the constants indicated buffered video lengths. The user retention probability of video k at the current chunk is defined as p_k , which is calculated as a posterior probability. We believe that it is necessary to adjust the upper threshold based on the user retention probability because the current video only needs less preload data when the user quits more easily.

If the video selector does not select any videos, the client will pause the download process. In our mechanism, we simply set the sleep time as a constant t_{sleep} .

3.3 Chunk Evaluator

The chunk evaluator evaluates each bitrate for every candidate chunk given by the video selector and finally downloads the best one. For the next to download video k and its bitrate b , we design a quality score to predict the expected reward as follows

$$Score(k, b) = p_k (w_1 \cdot Q(k, b) - w_3 \cdot S(k, b) - w_2 \cdot R(k, b) - \beta w_4 \cdot B(k, b)) \quad (6)$$

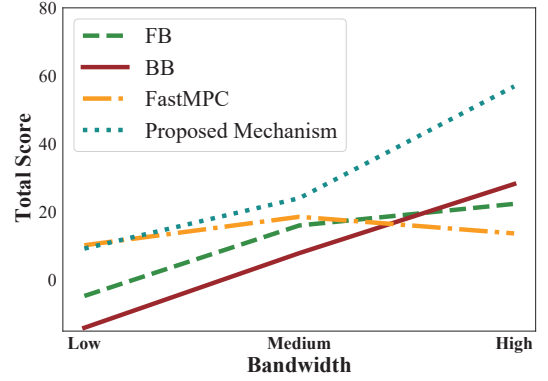
where $Q(\cdot)$, $S(\cdot)$, $R(\cdot)$, and $B(\cdot)$ denote the bitrate benefit, the smooth penalty, the rebuffer penalty, and the bandwidth usage, respectively. β is a non-negative balance parameter less than 1. Next we will introduce these functions and the balance parameter.

Smooth penalty. In Eq. 6, the video quality score $Q(k, b)$ and bandwidth usage $B(k, b)$ can be calculated directly without estimation. The smooth penalty $S(k, b)$ should be reflected in future download decisions instead of a one-step download decision. Based on this observation, we set the smooth penalty as follows

$$S(k, b) = \frac{b}{B_i \cdot buf_k} \cdot (b - b_k^{(last)}), \quad (7)$$

where buf_k is the unplayed buffered length of video k at current time and $b_k^{(last)}$ denote the last chunk's bitrate of video k .

Rebuffer penalty. Since the short video application can not know when the user stops watching the current video, the rebuffer time $R(k, b)$ should estimate according to the user retention model.

**Figure 4: Total score in different network conditions.**

We define the probability that the user will watch video k as $P_k = p_k \cdot (1 - \sum_{i=0}^{k-1} P_i)$. The rebuffer time can be estimated as follows

$$R(k, b) = \sum_{i=0}^k P_i \cdot \max(0, \frac{b}{B_i} - buf_i). \quad (8)$$

Balance parameter. β is a non-negative balance parameter less than 1 to adapt to the network conditions. When the bandwidth is relatively abundant, β is closer to 0, otherwise, β is closer to 1. β can be updated at each step as follows:

$$\beta = \begin{cases} 1 & B_i < B_{min}; \\ 1 - \frac{B_i - B_{min}}{B_{max} - B_{min}} & B_{min} < B_i < B_{max}; \\ 0 & B_i > B_{max}; \end{cases} \quad (9)$$

where B_{min} and B_{max} denote an estimation value of lower and higher bandwidth, respectively.

We finally select the combination (k^*, b^*) with the best expected reward $Score(k^*, b^*)$.

4 EVALUATION

We next evaluate the performance of the proposed download mechanism through simulations from multiple perspectives and compare it with three baseline algorithms.

4.1 Methodology

Datasets. We use the datasets provided by the "Short Video Streaming" challenge organizing community², which contain 7 video chunk traces with user retention models for each video and 60 network traces (20 traces in low, medium, high bandwidth, respectively). In our evaluation, we sampled 20 users using the user retention models. Since these network traces are generated using hidden Markov chain, we added 30 extra network traces (10 traces in low, medium, and high bandwidth, respectively) collected in the real world. Table 1 shows the feature of our bandwidth datasets.

Simulator settings. We use the simulator provided by the "Short Video Streaming" challenge organizer³ which can simulate the process of a short video application downloading video chunks periodically with a specific download mechanism. Note that in this simulator, the application can not obtain the user retention time

²<https://github.com/AltransCompetition/Short-Video-Streaming-Challenge/tree/main/data>

³<https://github.com/AltransCompetition/Short-Video-Streaming-Challenge>

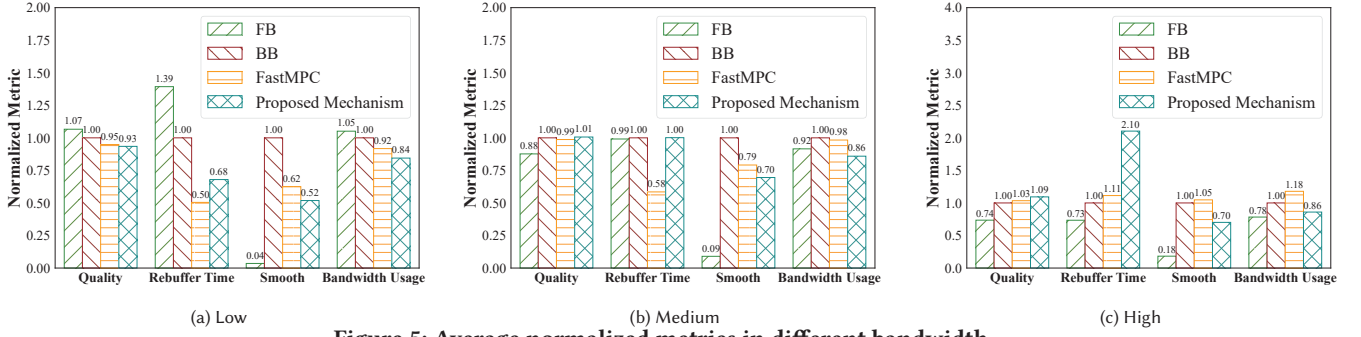


Figure 5: Average normalized metrics in different bandwidth.

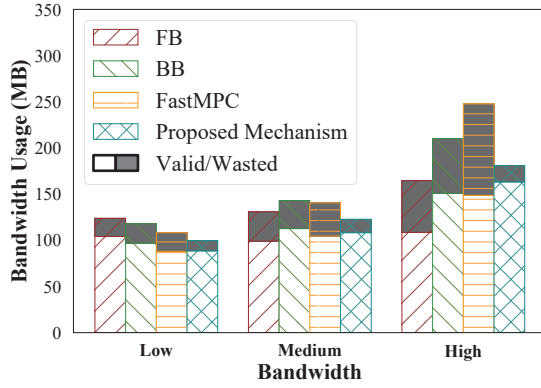


Figure 6: Bandwidth efficiency in various network conditions.

and future network bandwidth, but it can use user retention models to predict user behaviors. We empirically set the parameters in our video download mechanism as in Table 2.

Baselines. We compare our download mechanism with three baseline bitrate selection methods, namely, 1) Fixed Bitrate (FB) – a method which always download videos using a fixed bitrate [2]; 2) Buffer-based Bitrate (BB) – a method which selects the video bitrate based on the unplayed buffered amount [3]; 3) FastMPC – a typical DASH-based adaptive bitrate algorithm [8]. Note that in these baseline methods, the client preloads the videos in the recommendation queue in order, which is called Fixed Preload (FP). The maximum of preloading size is 4 chunks for each video.

4.2 Performance Comparison

Performance improvement. We first compare the quality score of the proposed mechanism with the baseline methods under various bandwidth conditions. Figure 4 shows that the proposed mechanism can improve the score under different bandwidth conditions. FastMPC is superior to other baseline methods under low and medium bandwidth since it can adapt bitrate based on bandwidth changes. FB and BB can improve the total score as bandwidth increases since they will select higher bitrate under higher bandwidth.

To understand the advantages of our method more clearly, we compare the QoE metrics (§2.1) and bandwidth usage of the proposed mechanism with baseline methods under different network bandwidths and plot the result in Figure 5. Note that we use the results of BB to normalize the performance metrics for the other

methods. The proposed mechanism prefers to maintain the video bitrate to ensure the smooth of the video playback and FastMPC prefers to select lower bitrate to avoid rebuffer. Under the medium and high bandwidth conditions, the proposed mechanism can obtain the highest video quality with more rebuffering time (the absolute value of rebuffering time for different methods is small).

Bandwidth efficiency. We compare the bandwidth efficiency of the proposed mechanism with that of the baseline methods by checking the valid and wasted bandwidth. Figure 6 shows that the proposed mechanism wastes the least bandwidth in all the cases while the other baseline methods waste more bandwidth as the available bandwidth increases because they download chunks without considering the user retention models. In our mechanism, the video selector selects video chunks based on the user retention models, the bandwidth estimations, and the video chunk sizes, which stop downloading when the user retention probability is too low. The evaluation results show that the proposed mechanism mitigate the conflicts between user QoE and network bandwidth usage, which outperforms baseline methods under various bandwidth conditions.

5 CONCLUSION

In this paper, we proposed a QoE-aware video download mechanism to optimize the user QoE and minimize the bandwidth usage in short video applications. We introduced the video download problem in short video applications. Further, we discussed the performance metrics and designed a video download control mechanism including three major components. To evaluate the performance of our design, we conducted extensive simulations and showed that our mechanism can improve the user QoE with less bandwidth waste compared to the other baseline methods.

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